

Stellar Physics

How a star spends its whole life
balancing two forces against each
other.

Advanced Higher Physics



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

LEARNING INTENTION

To use the relationships that describe a star's power and brightness, read a Hertzsprung-Russell diagram, and describe how a star forms, lives and ends.

By the end of this topic...

- 1 I can use $P/A = \sigma T^4$, $L = 4\pi r^2 \sigma T^4$ and $b = L/(4\pi d^2)$ to solve problems about stars.
- 2 I can classify a star from its position on a Hertzsprung-Russell diagram, and predict its colour.
- 3 I can describe how a star forms, and state the three stages of the proton-proton chain.
- 4 I can describe how a star leaves the main sequence, and explain what its mass determines.

Daily Review: Gravitation. Close your notes and write your answers.



- 1 State the definition of escape velocity.
- 2 Write the relationship for gravitational potential energy.
- 3 State Newton's universal law of gravitation.
- 4 State the definition of gravitational field strength.
- 5 Sketch the gravitational field around two equal masses.

Every star in this photo is a different colour. What does that tell you?



PART 1

How hot? How big?

Three relationships, and the light already reached us.

By the end of this part...



1

I can use $P/A = \sigma T^4$ to find the power radiated per unit area of a star.

2

I can use $L = 4\pi r^2 \sigma T^4$ to find a star's luminosity, radius or surface temperature.

3

I can use $b = L/(4\pi d^2)$ to find a star's apparent brightness.

Power per unit area



$$\frac{P}{A} = \sigma T^4$$

P

power (W)

A

surface area
(m²)

σ

a constant
(W m⁻² K⁻⁴)

T

surface
temperature
(K)

σ is the Stefan-Boltzmann constant. Every star is treated as a black body.

Power per unit area: a star is a black body



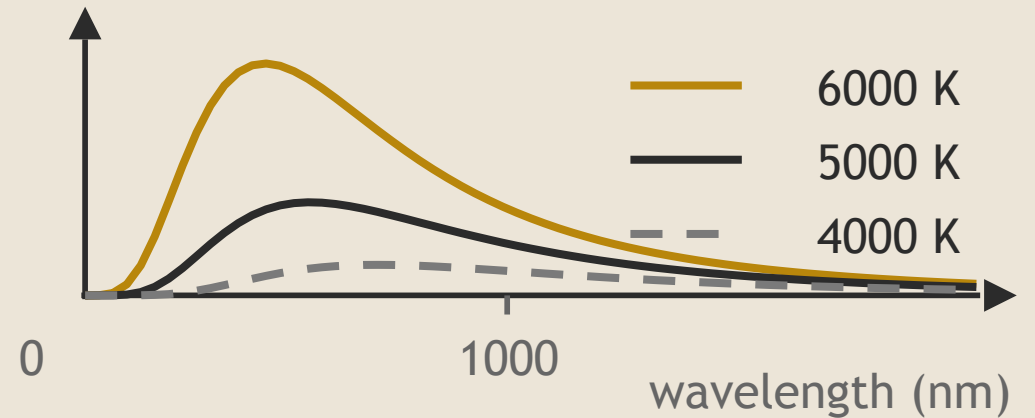
$$\frac{P}{A} = \sigma T^4$$

The power radiated per square metre of a star's surface.

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

Double T and P/A rises $\times 16$.

intensity (arbitrary units)



Cool Betelgeuse glows orange; hot Rigel glows blue-white.

Worked example 1 – power per unit area

A star has a surface temperature of 6000 K.

Calculate the power radiated per unit area from its surface.

$$T = 6000 \text{ K}$$

$$P/A = ?$$



$$\frac{P}{A} = \sigma T^4$$

$$\frac{P}{A} = 5.67 \times 10^{-8} \times 6000^4$$

$$P/A = 7.35 \times 10^7 \text{ W m}^{-2}$$

Luminosity



$$L = 4\pi r^2 \sigma T^4$$

L

luminosity (W)

r

radius of the
star (m)

σ

a constant
(W m⁻² K⁻⁴)

T

surface
temperature
(K)

$4\pi r^2$ is the surface area of the star, so $L = P/A \times A$.

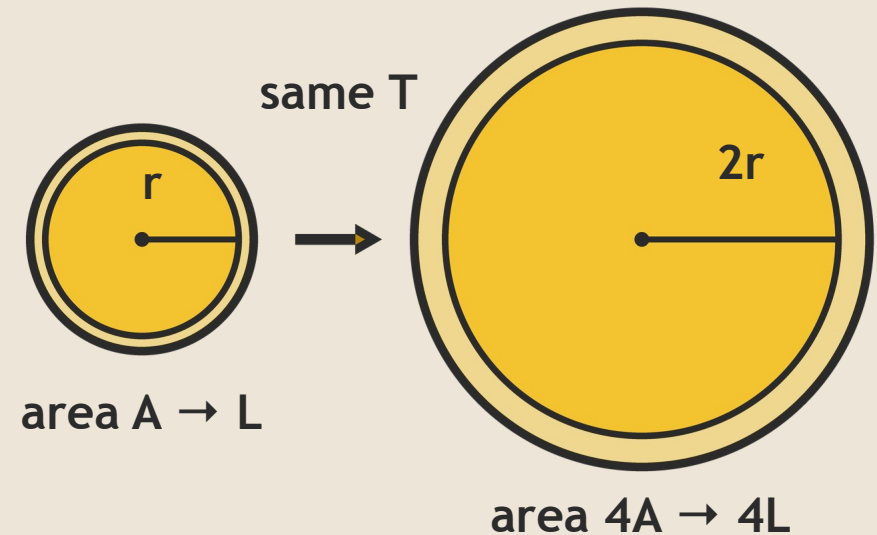
Luminosity: the total power in every direction



$$L = 4\pi r^2 \sigma T^4$$

The TOTAL power radiated, over the star's whole surface area.

A star can be luminous by being hot, by being big, or both.



Worked example 2 – luminosity

A star has a radius of $8.0 \times 10^8 \text{ m}$ and a surface temperature of 5000 K . Calculate its luminosity.

$$r = 8.0 \times 10^8 \text{ m}$$

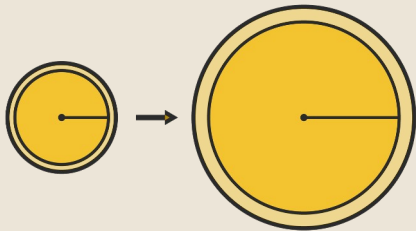
$$T = 5000 \text{ K}$$

$$L = ?$$

$$L = 4\pi r^2 \sigma T^4$$

$$L = 4\pi(8.0 \times 10^8)^2 \times 5.67 \times 10^{-8} \times 5000^4$$

$$L = 2.85 \times 10^{26} \text{ W}$$



Apparent brightness



$$b = \frac{L}{4\pi d^2}$$

b

apparent
brightness (W
m⁻²)

L

luminosity (W)

d

distance to the
star (m)

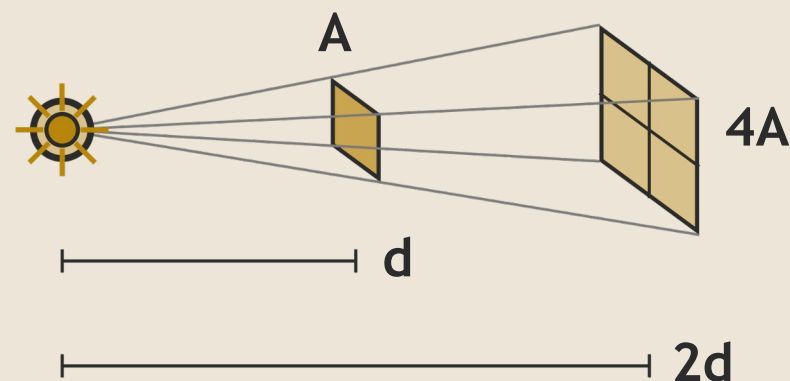
Convert light-years or parsecs to metres first.

Apparent brightness fades with distance



$$b = \frac{L}{4\pi d^2}$$

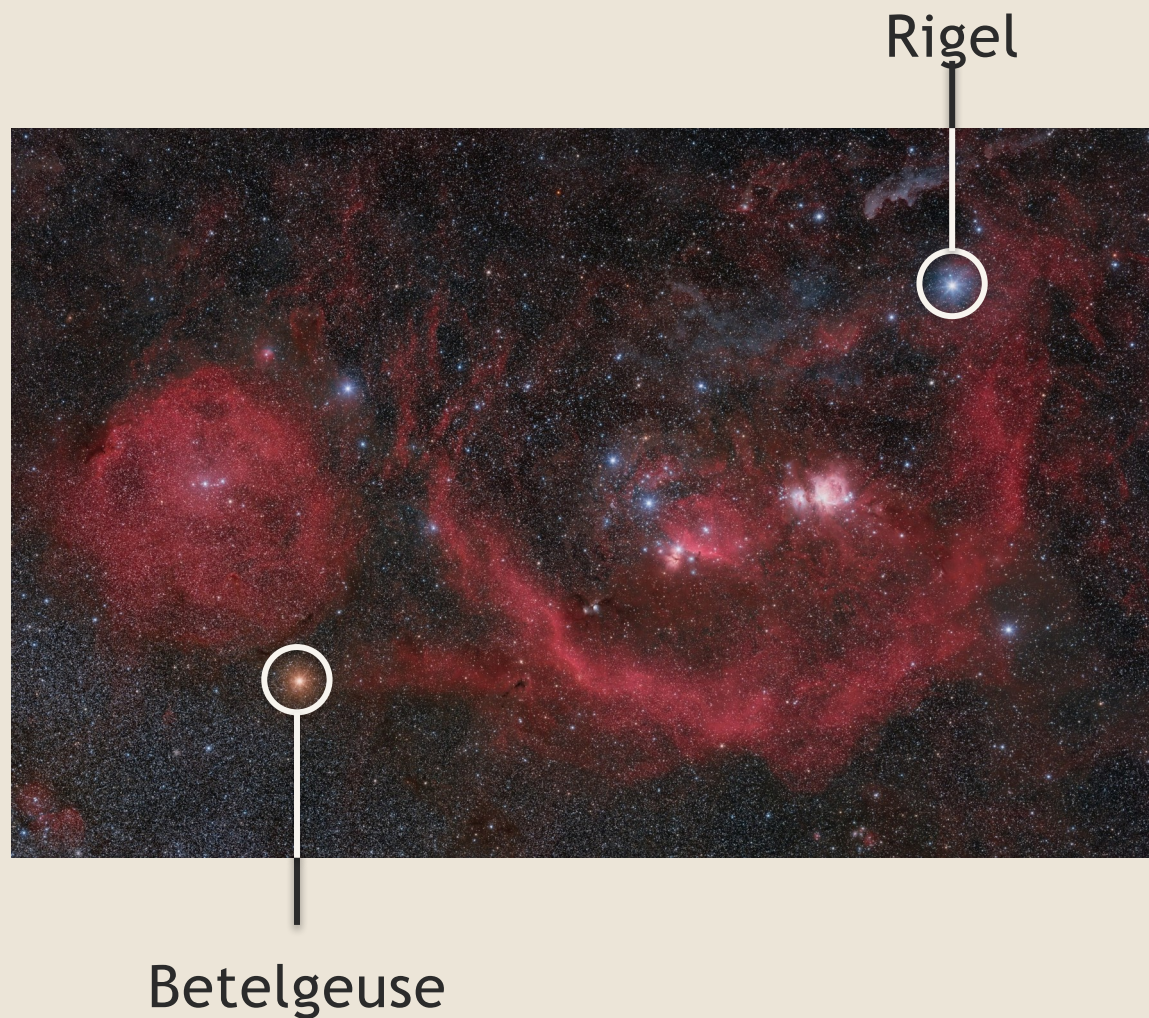
The same luminosity,
spread over a sphere of
radius d .



Same brightness, different stars



Two stars can look equally bright for very different reasons — one dim and close, one luminous and far. From Earth they look alike — but b alone cannot give you L .



Worked example 3 – apparent brightness – 2023 Q6(d)

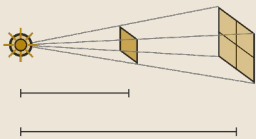
Delta Ursae Majoris has a luminosity of $5.46 \times 10^{27} \text{ W}$ and is **24.7 parsecs** from Earth (1 parsec = 3.26 ly).

Calculate its apparent brightness.

$$L = 5.46 \times 10^{27} \text{ W}$$

$$d = 24.7 \text{ pc}$$

$$b = ?$$



$$d = 24.7 \times 3.26 = 80.5 \text{ ly}$$

$$d = 80.5 \times 9.46 \times 10^{15} = 7.62 \times 10^{17} \text{ m}$$

$$b = \frac{L}{4\pi d^2}$$

$$b = 7.48 \times 10^{-10} \text{ W m}^{-2}$$

Look back — by the end of this part...



1

I can use $P/A = \sigma T^4$ to find the power radiated per unit area of a star.

2

I can use $L = 4\pi r^2 \sigma T^4$ to find a star's luminosity, radius or surface temperature.

3

I can use $b = L/(4\pi d^2)$ to find a star's apparent brightness.

Daily Review: Part 1. Close your notes and write your answers.



- 1 Write the relationship for power radiated per unit area.
- 2 Write the relationship for the luminosity of a star.
- 3 Write the relationship for the apparent brightness of a star.
- 4 A star's temperature doubles. By what factor does its power per unit area increase?
- 5 What assumption do all three relationships make about a star?

Astronomers plot clusters like this on a graph. What could a point on that graph tell you?



PART 2

Why plot stars on a graph?

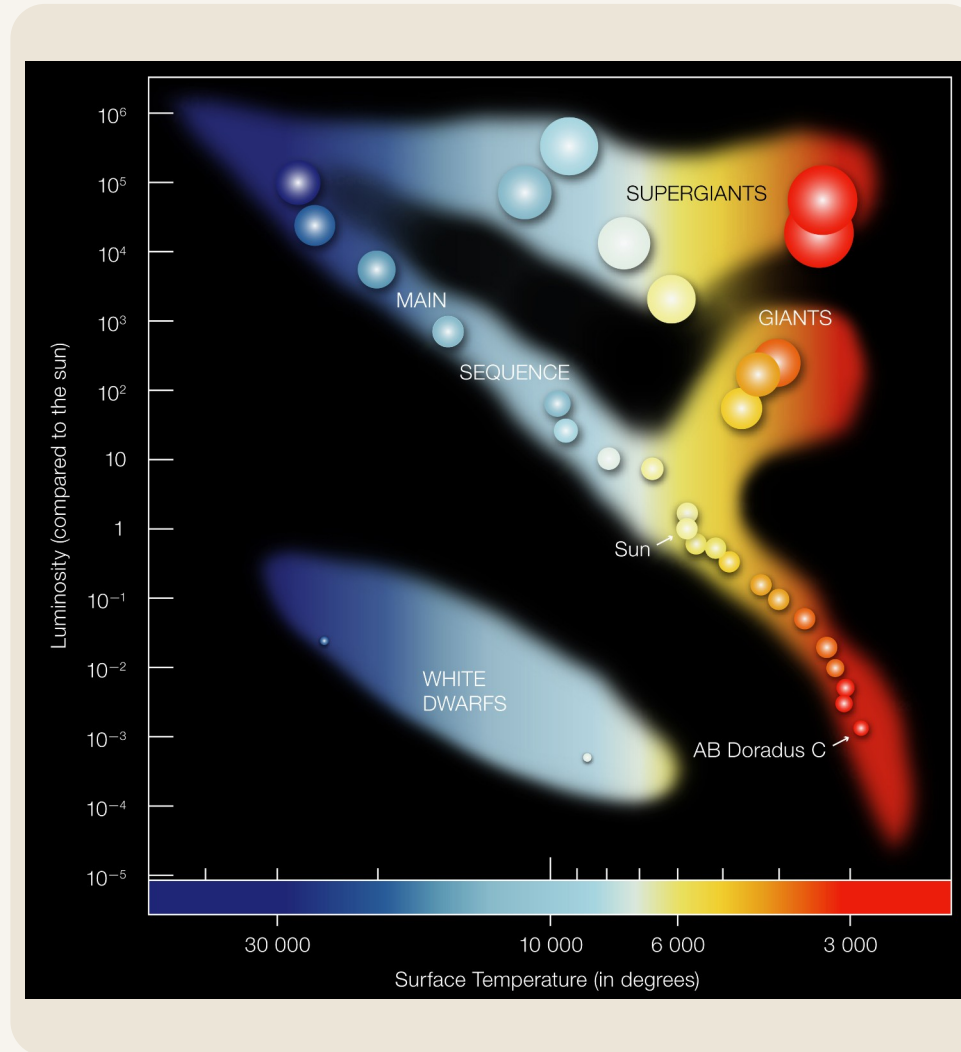
Position says what kind of star it is.

By the end of this part...



- 1 I can describe what a Hertzsprung-Russell diagram represents.
- 2 I can classify a star as main sequence, giant, supergiant or white dwarf from its position.
- 3 I can predict a star's colour from its position on the diagram.

The H-R diagram: hot on the left, log scale up



Reading colour and position

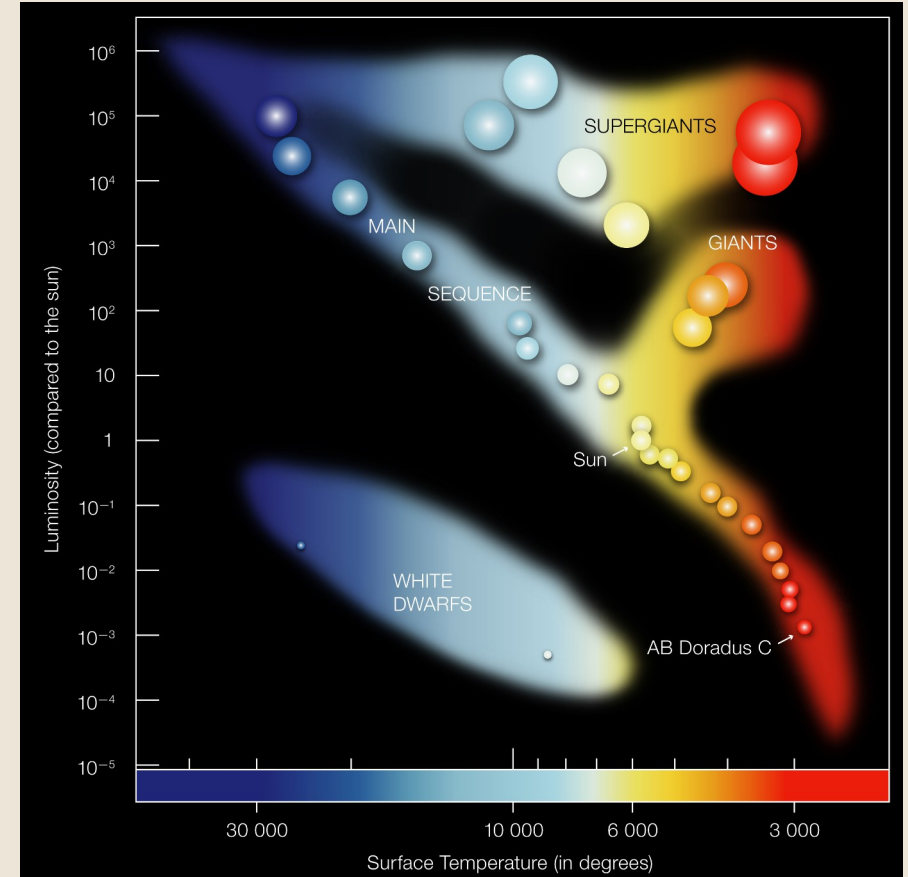


Colour

Hot, blue-white stars sit on the LEFT. Cool, red stars sit on the RIGHT.

Off the main sequence

Giants and supergiants are cool but luminous, so huge. White dwarfs are hot but faint, so tiny.



Worked example 4 — reading the diagram — 2023 Q6(e)

Epsilon Ursae Majoris has a luminosity about two orders of magnitude greater than the Sun's, and a surface temperature of **about 9000 K**. Mark its position on the diagram.

$$L \approx 100 \times L_{\text{Sun}}$$

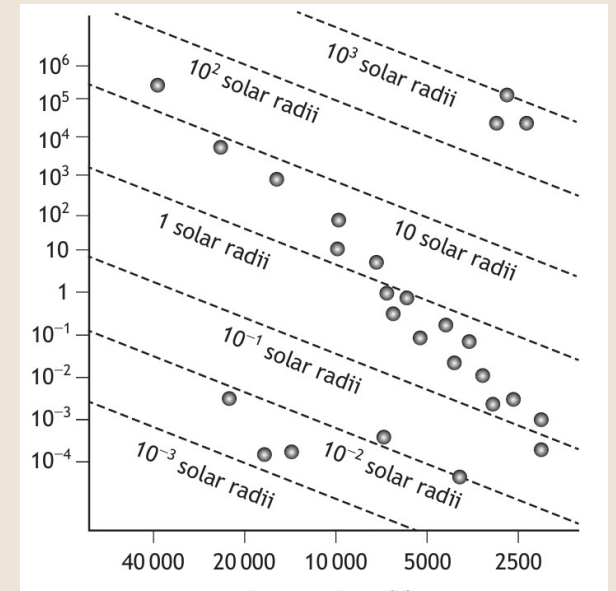
$$T \approx 9000 \text{ K}$$

position = ?

2 orders of magnitude = $\times 100$

plot at $L \approx 10^2$, $T \approx 9000 \text{ K}$

**just above the main
sequence — in the giant
region**



Look back — by the end of this part...



- 1 I can describe what a Hertzsprung-Russell diagram represents.
- 2 I can classify a star as main sequence, giant, supergiant or white dwarf from its position.
- 3 I can predict a star's colour from its position on the diagram.

Daily Review: Part 2. Close your notes and write your answers.



- 1 What does a Hertzsprung-Russell diagram plot on each axis?
- 2 Where on the diagram do main-sequence stars lie?
- 3 Where do giants and supergiants lie, compared to the main sequence?
- 4 Where do white dwarfs lie, and why is that surprising given how hot they are?
- 5 What colour is a star with a high surface temperature?

Somewhere in this photo, a star is being born right now.
How?



PART 3

What builds a star?

Gravity pulling in, until fusion pushes back.

By the end of this part...



- 1 I can describe how a star forms from an interstellar cloud.
- 2 I can explain why a main-sequence star is stable.
- 3 I can state the three stages of the proton-proton chain.
- 4 I can describe the net effect of the proton-proton chain.

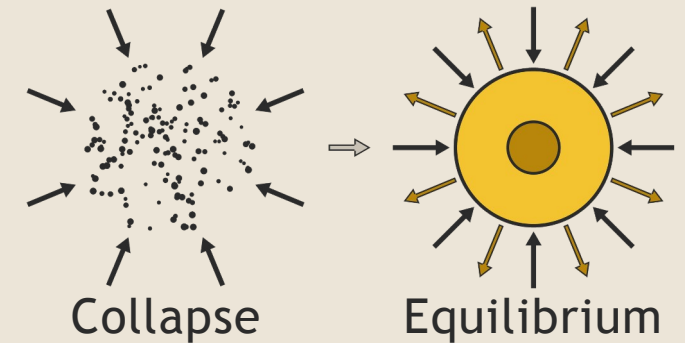
Life of a star 1 - A star is born



An interstellar gas cloud collapses under its own gravity.

Core compression drives temperatures to **15 million kelvin**.

This extreme environment triggers the **nuclear fusion of hydrogen into helium**, igniting the necessary outward pressure to stop the collapse.



Black: gravity. Gold: thermal pressure.

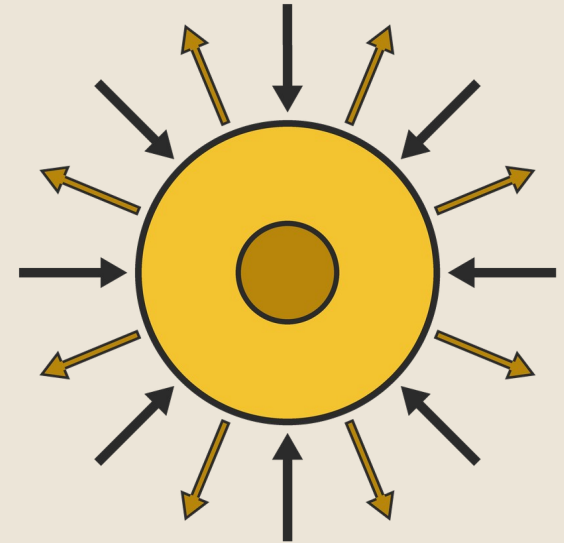
Life of a star 2 - Hydrostatic equilibrium



Inward gravitational pull perfectly balances outward pressure.

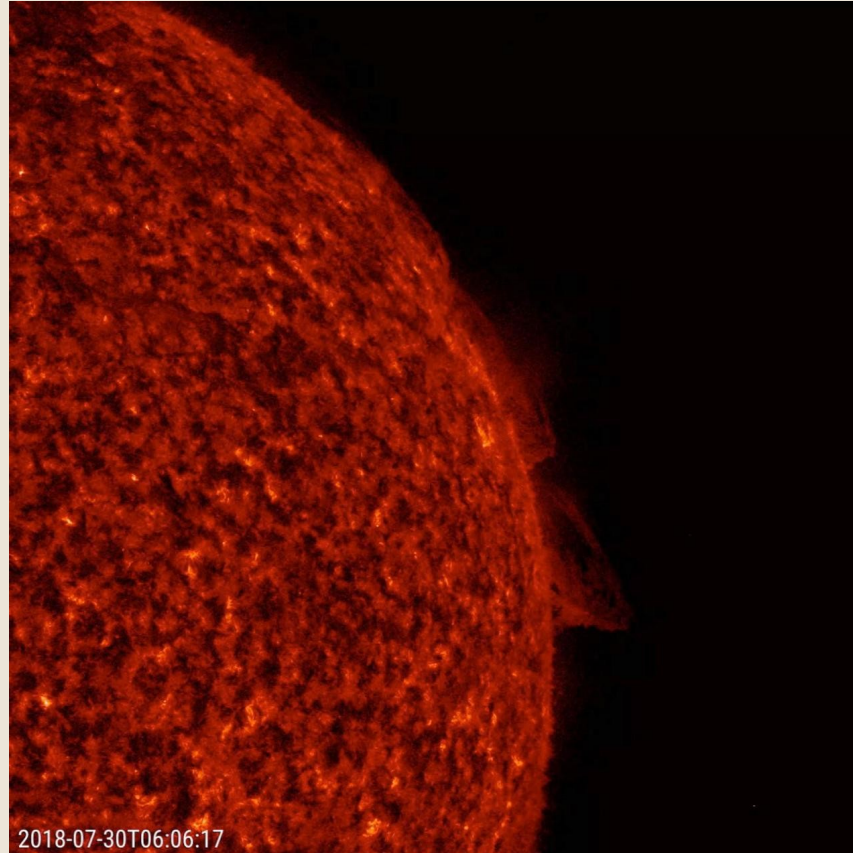
Powered entirely by the steady **fusion of hydrogen into helium** in the core.

This stable state preserves the Sun's size for a roughly 10-billion-year lifespan.



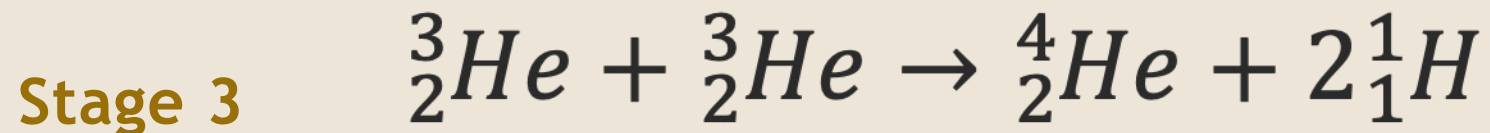
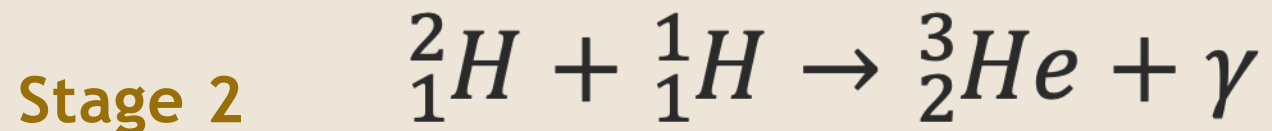
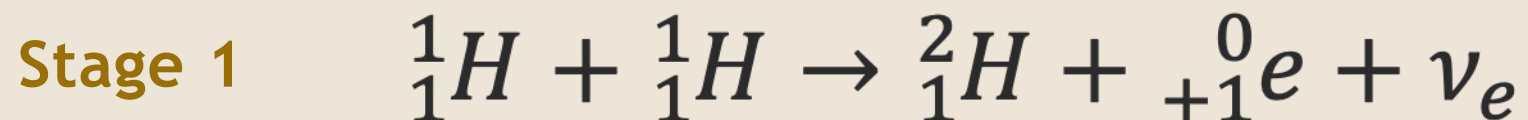
Black: gravity. Gold: thermal pressure.

The Sun — a main-sequence star right now



Prominences on the Sun's surface (SDO).

The proton-proton chain



Net effect: 4 protons $\rightarrow$ 1 helium-4 nucleus.

Worked example 5 – the real p-p chain – 2024 Q6(b)

Figure 6A shows one possible p-p chain. State the name of the particles represented by ν_e and ${}^0_{+1}e$, and suggest why deuterium ${}^2_1\text{H}$ does not build up in the star.

$\nu_e = ?$

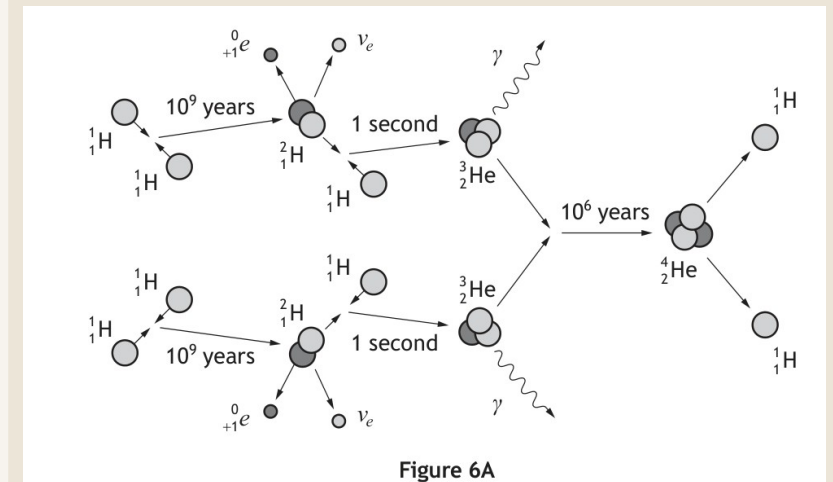
${}^0_{+1}e = ?$

${}^2_1\text{H}$ build-up = ?

ν_e = (electron) neutrino

${}^0_{+1}e$ = positron

deuterium exists for only 1 second before fusing again



Look back — by the end of this part...



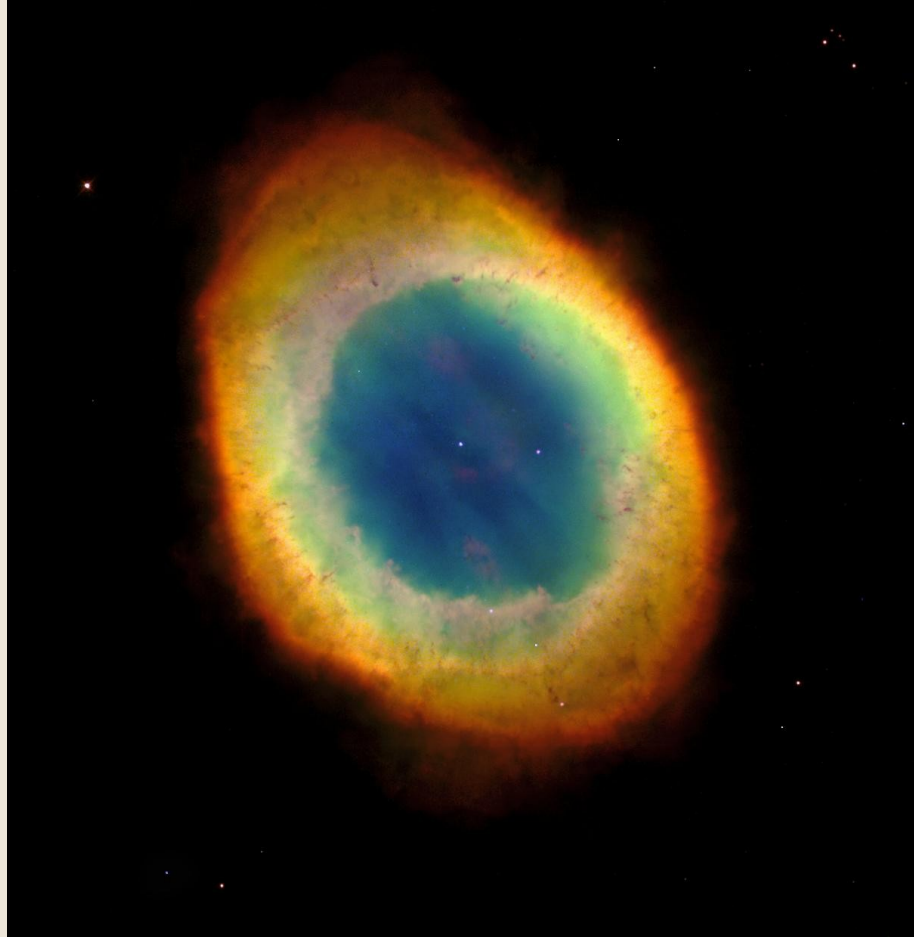
- 1 I can describe how a star forms from an interstellar cloud.
- 2 I can explain why a main-sequence star is stable.
- 3 I can state the three stages of the proton-proton chain.
- 4 I can describe the net effect of the proton-proton chain.

Daily Review: Part 3. Close your notes and write your answers.



- 1 What two forces are balanced in a stable, forming star?
- 2 Write the first stage of the proton-proton chain.
- 3 What particles are released at each stage of the proton-proton chain?
- 4 What is the net effect of the proton-proton chain?
- 5 Where in a star does hydrogen fusion happen?

This glowing ring used to be a star. What happened to it?



PART 4

What decides a star's fate?

One number: mass.

By the end of this part...



1

I can describe what happens when a star like the Sun leaves the main sequence.

2

I can describe how a star like the Sun ends as a white dwarf.

3

I can state that a star's mass determines its lifetime and its eventual fate.

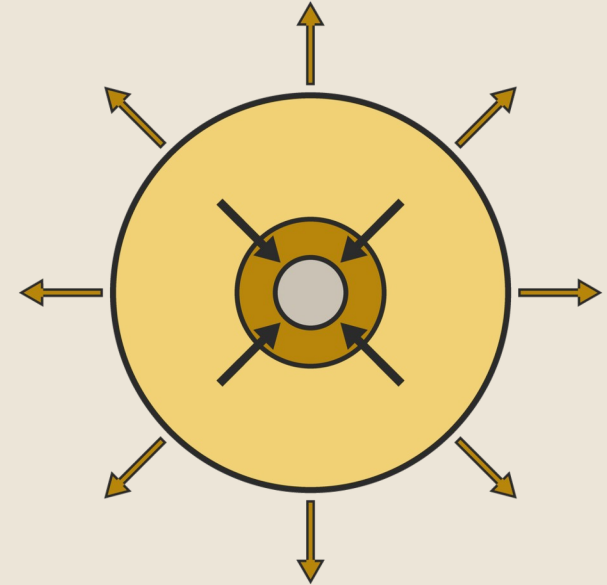
Life of a star 3 - The road to death



Hydrogen runs out in the core, core fusion stops, and gravity crushes the core inward.

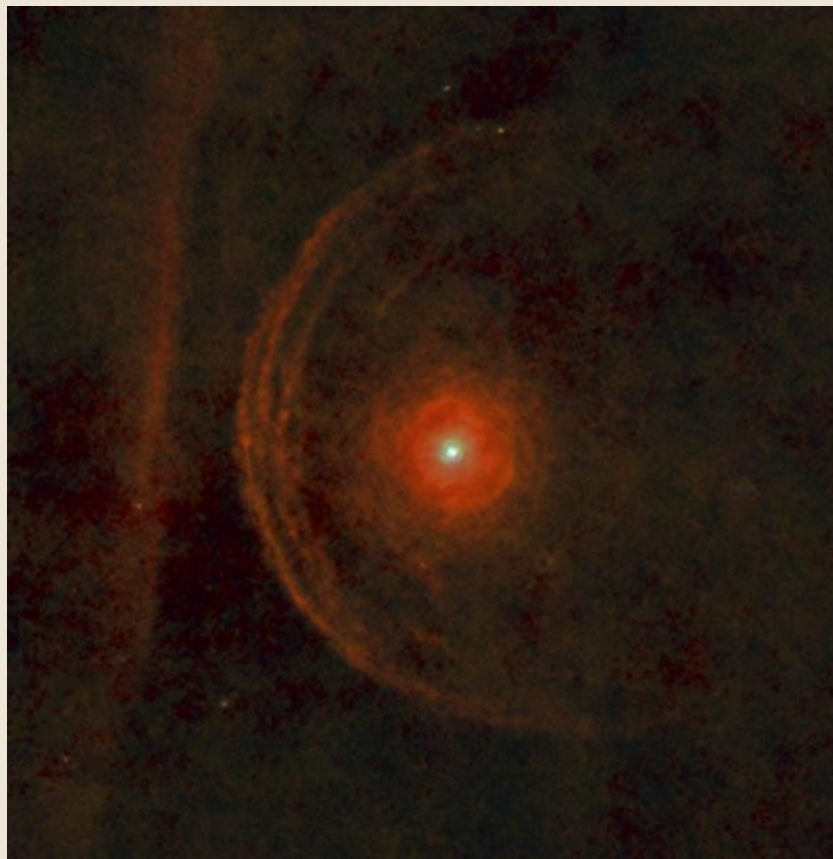
The **shell of hydrogen** around the core heats up until it fuses.

The extra pressure pushes the surface outward, and it cools: the star becomes a **red giant**.



Grey: core. Gold:
the fusing shell.

Betelgeuse — a real red supergiant

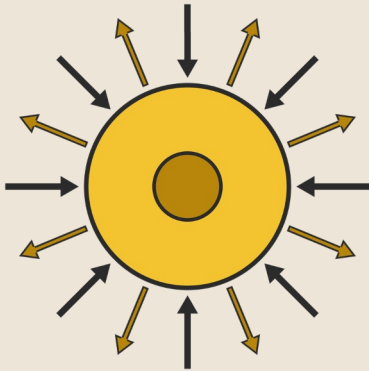


Shells of gas around Betelgeuse (Herschel).

Worked example 6 — becoming a red giant — 2023 Q6(c)

Explain, in terms of thermal pressure and gravitational forces, what happens to the radius of the Sun as it becomes a red giant.

radius = ?



radius increases —

thermal pressure $>$ gravity

stops increasing when they balance

Life of a star 4 - The end of the road

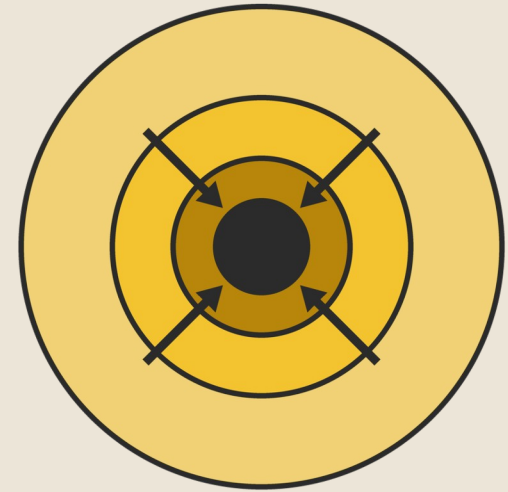


The core shrinks until it is hot enough to **fuse helium into carbon**.

A star like the Sun lacks the mass to compress its core any further.

It cannot reach the **600 million kelvin** needed to fuse carbon.

With no new fusion, the outward thermal pressure stops for good.



Black: carbon core.
Gold: helium.

Life of a star 5 - A new hope

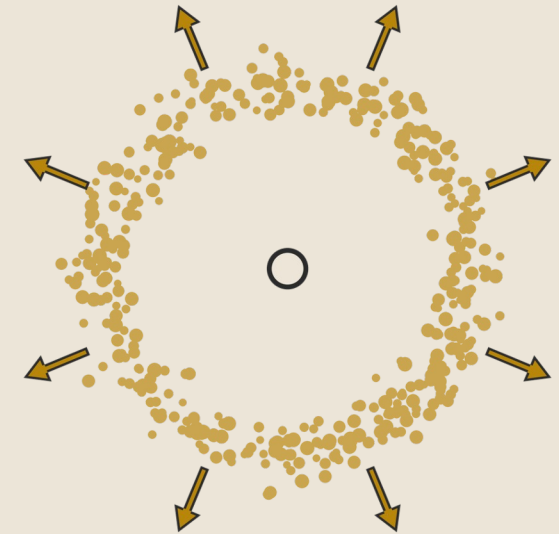


Ejection: the Sun gently sheds its outer gas layers into space.

Nebula glow: this expanding shell forms a glowing **planetary nebula**.

Core remnant: the exposed, dead core stays at the centre.

Final state: this dense, cooling ember is a **white dwarf**.



Centre: the white dwarf.

Mass decides the fate

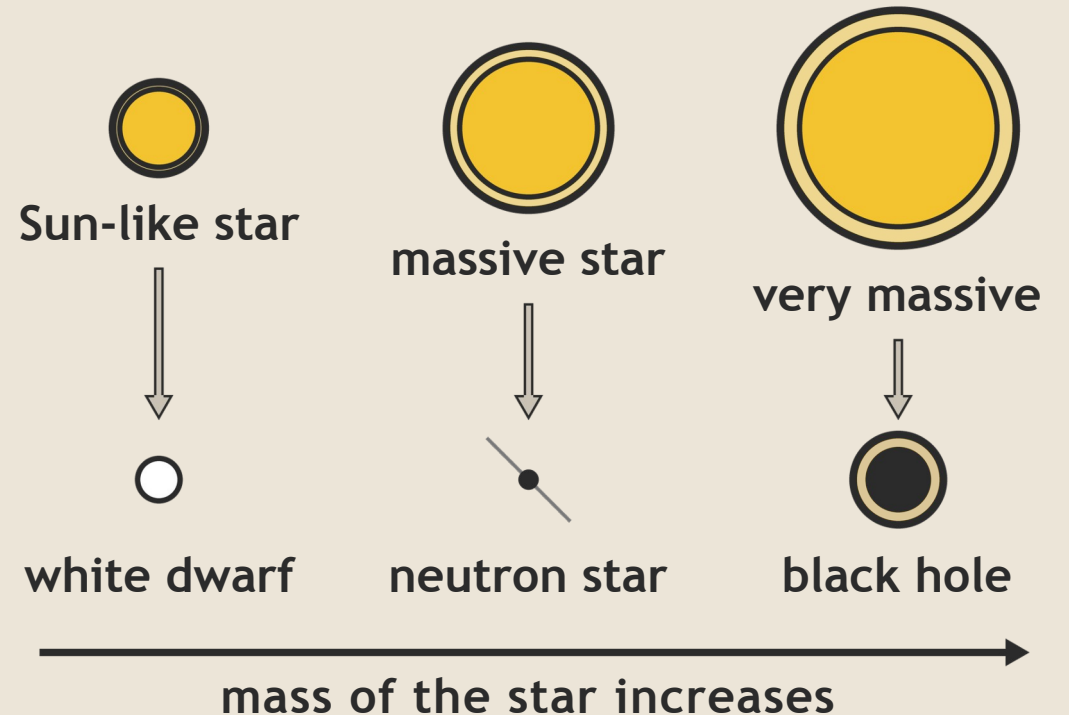


Heavier stars go further

They fuse helium to carbon, then neon, oxygen, silicon, and finally iron.

Three endings

Mass determines lifetime, and the ending: white dwarf, neutron star, or black hole.



The Crab Nebula — a real supernova remnant



A neutron star lies at the heart of this nebula.

Look back — by the end of this part...



1

I can describe what happens when a star like the Sun leaves the main sequence.

2

I can describe how a star like the Sun ends as a white dwarf.

3

I can state that a star's mass determines its lifetime and its eventual fate.

Stellar physics in every diet since 2016

Diet	Question	What it asked
'16,17,19,20,22	varies	Calculation
2018	Q6, Q8	Apparent brightness; p-p chain
2023	Q6	Full topic – worked in class above
2024	Q6	Full topic – worked in class above
2025	Q3, Q5	Evolution; Sirius, Rigel, Betelgeuse



Scholar Topic 7 test

Read BrightRed pp. 46-58.

Past papers: 2016-2020, 2022, 2025.



Scholar

Errors that cost marks every year

- Read carefully which of P/A , L or b is being asked for.
- Convert parsecs and light-years to metres before substituting.
- On an H-R diagram, hot is on the LEFT, not the right.
- A white dwarf is hot but faint – because it is small, not cool.
- Black-hole mechanics is General Relativity, not this topic.

Check yourself, without your notes

- 1 Use all three relationships, given any two of L , b , d , r , T .
- 2 Sketch an H-R diagram — main sequence, giants, supergiants, white dwarfs.
- 3 Write all three stages of the proton-proton chain.
- 4 Explain why a star leaves the main sequence for a giant.
- 5 State a star's three possible fates, and what decides between them.

Experiments you could try

- Use a spectrometer to compare colour against temperature.
- Plot stars from a catalogue onto your own H-R diagram.
- Use Stellarium to look up a named star's temperature.
- Model gravitational contraction with a deflating balloon.
- Research a star's class and place it on the H-R diagram.

Course specification – 1 of 7

- Use of appropriate relationships to solve problems relating to luminosity, apparent brightness b , distance between the observer and the star, power per unit area, stellar radius, and stellar surface temperature.
- (Using the assumption that stars behave as black bodies.)

Course specification – 2 of 7

- Knowledge that stars are formed in interstellar clouds when gravitational forces overcome thermal pressure and cause a molecular cloud to contract until the core becomes hot enough to sustain nuclear fusion,
- which then provides a thermal pressure that balances the gravitational force.

Course specification – 3 of 7

- Knowledge of the stages in the proton-proton chain (p-p chain) in stellar fusion reactions which convert hydrogen to helium.
- Knowledge that Hertzsprung-Russell (H-R) diagrams are a representation of the classification of stars.
- Classification of stars and position in Hertzsprung-Russell (H-R) diagrams, including main sequence, giant, supergiant, and white dwarf.

Course specification – 4 of 7

- Use of Hertzsprung-Russell (H-R) diagrams to determine stellar properties, including prediction of colour of stars from their position in an H-R diagram.
- Knowledge that the fusion of hydrogen occurs in the core of stars in the main sequence of a Hertzsprung-Russell (H-R) diagram.
- Knowledge that, in a star like the Sun, the core shrinks and will become hot enough for the helium in the core to begin fusion.

Course specification – 5 of 7

- Knowledge that hydrogen fusion in the core of a star supplies the energy that maintains the star's outward thermal pressure to balance inward gravitational forces. When the hydrogen in the core becomes depleted, nuclear fusion in the core ceases.
- The gas surrounding the core, however, will still contain hydrogen. Gravitational forces cause both the core, and the surrounding shell of hydrogen, to shrink.

Course specification – 6 of 7

- In a star like the Sun, the hydrogen shell becomes hot enough for hydrogen fusion in the shell of the star. This leads to an increase in pressure which pushes the surface of the star outwards, causing it to cool.
- At this stage, the star will be in the giant or supergiant regions of a Hertzsprung-Russell (H-R) diagram.

Course specification – 7 of 7

- Knowledge that the mass of a star determines its lifetime.
- Knowledge that every star ultimately becomes a white dwarf, a neutron star or a black hole. The mass of the star determines its eventual fate.

Credits (1 of 3)

Title slide: Photo: Orion: Giuseppe Donatiello, CC0

Power per unit area: a star is a black body:

Photo: Orion: Giuseppe Donatiello, CC0

Worked example 1 — power per unit area:

Sun: NASA/GSFC/SDO. © Qualifications Scotland (SQA)

Worked example 2 — luminosity: ©

Qualifications Scotland (SQA)

Apparent brightness fades with distance:

Photo: Orion: Giuseppe Donatiello, CC0

Worked example 3 — apparent brightness — 2023 Q6(d): 2023 Q6(d). © Qualifications Scotland (SQA)

The H-R diagram: hot on the left, log scale up: H-R diagram: ESO, CC BY 4.0

Reading colour and position: H-R diagram: ESO, CC BY 4.0

Worked example 4 — reading the diagram — 2023 Q6(e): 2023 Q6(e), Figure 6C. © Qualifications Scotland (SQA)

Credits (2 of 3)

Worked example 5 – the real p-p chain –

2024 Q6(b): 2024 Q6(b), Figure 6A. ©

Qualifications Scotland (SQA)

The Sun – a main-sequence star right now:

Photo: NASA/GSFC/Solar Dynamics Observatory

Betelgeuse – a real red supergiant: Photo:

ESA/Herschel/PACS/L. Decin et al.

Worked example 6 – becoming a red giant –

2023 Q6(c): 2023 Q6(c). © Qualifications

Scotland (SQA)

The Crab Nebula – a real supernova remnant:

Photo: NASA

Stellar physics in every diet since 2016: ©

Qualifications Scotland (SQA)

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Part 1 photo question: Photo: NASA

Credits (3 of 3)

Part 2 photo question: Photo: NASA/JPL/Space Science Institute

Part 3 photo question: Photo: NASA

Part 4 photo question: Photo: NASA/JPL-Caltech/ESA, the Hubble Heritage Team STScI/AURA

Same brightness, different stars: Photo: Orion: Giuseppe Donatiello, CC0